

Tech & Teach

Digital Chip Design Program

RTL Design

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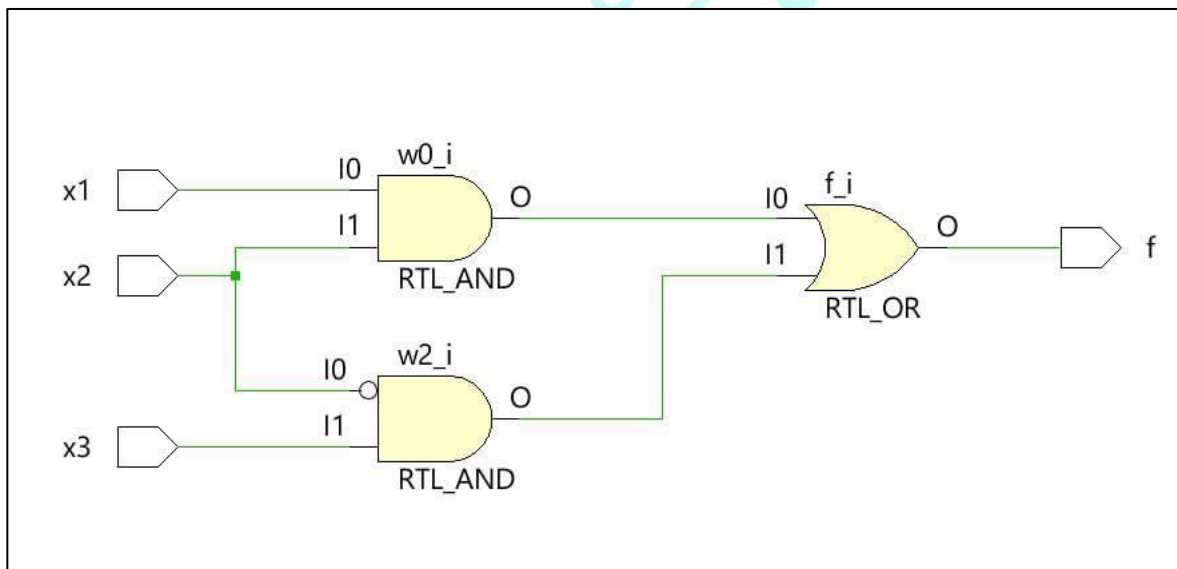
Task #2

30/6/2026

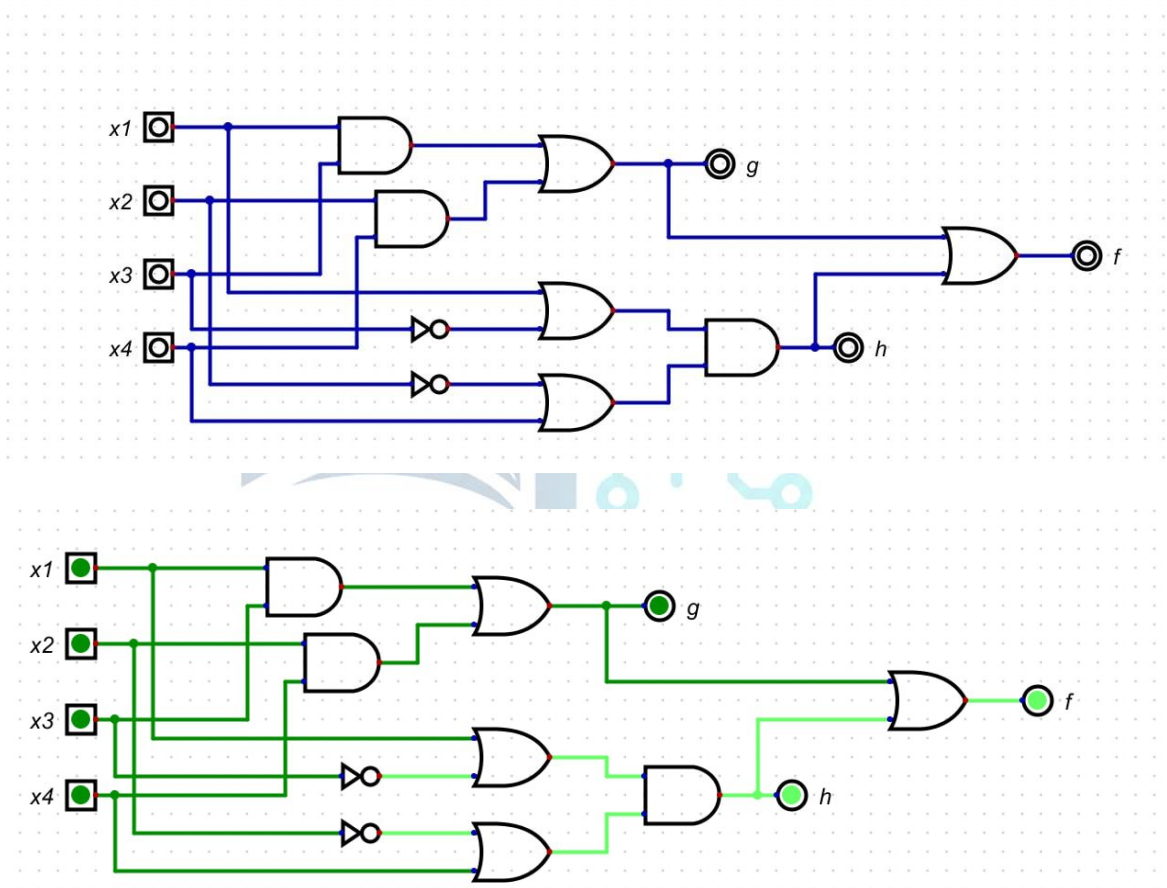
Data flow

Task #2.1

```
module task_2(  
  input logic x1,  
  input logic x2,  
  input logic x3,  
  output logic f  
);  
  assign f = ((x1&x2) | (~x2&x3));  
endmodule
```



Task #2.2



Task #2.3

```

module task2_3(
input logic a3,b3,a2,b2,a1,b1,a0,b0,
output logic AeqB, AgtB, AltB

);

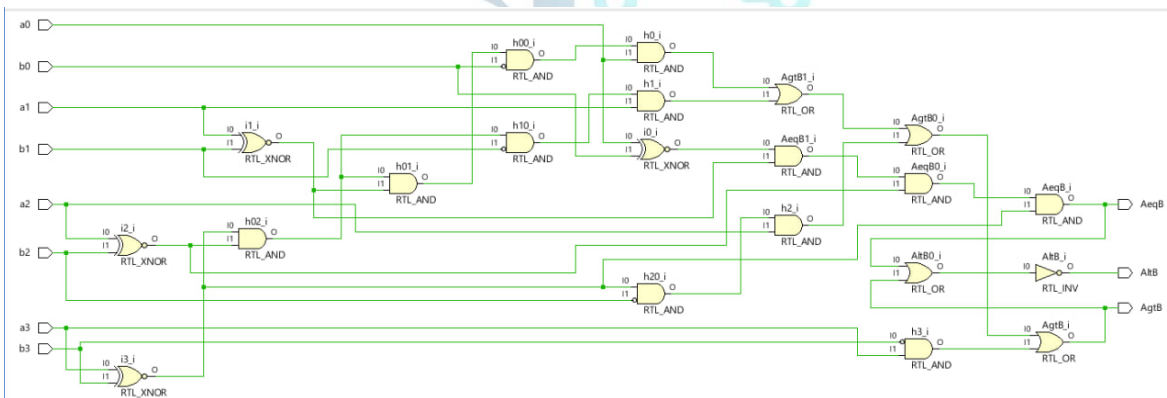
logic i3,i2,i1,i0;
logic h0,h1,h2,h3;

assign i3= (a3 ^ b3);
assign i2= (a2 ^ b2);
assign i1= (a1 ^ b1);
assign i0= (a0 ^ b0);
assign h3= (~b3 & a3);
assign h2= (i3 & ~b2 & a2);
assign h1= (i3 & i2 & ~b1 & a1);
assign h0= (i3 & i2 & i1 & ~b0 & a0);

assign AeqB = (i0 & i1 & i2 & i3);
assign AgtB = (h0 | h1 | h2 | h3);
assign AltB = ~(AeqB | AgtB);

endmodule

```



Work distribution

Section	ROAA	RAND	LANA	LARA	RAWAN	RENAD
Report					✓	✓
Task 1.1			✓	✓		
Task 1.2	✓	✓				
Task 1.3	✓	✓	✓	✓	✓	✓